A5112

MIL-STD-1312-1
19 OCTOBER 1984
SUPERSEDING
MIL-STD-1312 (In part)
31 MAY 1967

MILITARY STANDARD

FASTENER TEST METHODS

METHOD 1,

SALT SPRAY

FSC 53GP

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:37Z
Check the source to verify that this is the current version before use.

DEPARTMENT OF DEFENSE
WASHINGTON, DC 20301

Fastener Test Methods, Method 1, Salt Spray

MIL-STD-1312-1

1. This Military Standard is approved for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD), Code 93, Lakehurst, NJ 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:37Z
Check the source to verify that this is the current version before use.

FOREWORD

This standard sets forth a standard test method to simulate highly humid and salt atmospheric conditions to determine a fastener's relative resistance to corrosion.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:37Z
Check the source to verify that this is the current version before use.

TABLE OF CONTENTS

Paragraph		Page
1.	SCOPE	1
1.1	Applicability	1
2.	REFERENCED DOCUMENTS	1
3.	DEFINITIONS	1
4.	GENERAL REQUIREMENTS	1
4.1	Test apparatus	1
4.1.1	Exposure chamber	1
4.1.2	Atomizers	2
4.1.3	Air supply	2
4.1.4	Salt solution	2
4.2	Test specimens	3
4.2.1	General	3
5.	DETAIL REQUIREMENTS	4
5.1	Test procedures	4
5.1.1	Location of specimens	4
5.2	Operating conditions	4
5.2.1	Temperature	4
5.2.2	Atomization	4
5.3	Length of test	5
5.4	Examination of specimens	5
5.5	Baking of specimens	5
6.	NOTES	5
6.1	Test report	5

TABLE

Table		Page
I	Air pressure and temperature range	2

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:37Z
Check the source to verify that this is the current version before use.

1. SCOPE

1.1 Applicability. This method covers a procedure for testing fasteners in salt water spray to determine relative resistance to corrosion for maintaining standards of quality, although it does not closely simulate any common conditions of service.

2. REFERENCED DOCUMENTS

Not applicable.

3. DEFINITIONS

Not applicable.

4. GENERAL REQUIREMENTS

4.1 Test apparatus. Test apparatus used in the salt spray test shall include the following:

a. Exposure chamber with racks for supporting specimens.

b. Salt solution reservoir.

c. Means for atomizing salt solution, including suitable nozzles and compressed air supply.

d. Means for heating the exposure chamber and controlling its temperature.

e. Means for cleaning the air and humidifying it at a temperature above that of the exposure chamber.

4.1.1 Exposure chamber. The chamber and all accessories shall be made of material such as glass, hard rubber, plastic, or suitably-coated wood that will not affect the corrosiveness of the fog. In addition, all parts that come in contact with test specimens shall be made of materials that will not cause galvanic or electrolytic corrosion. The chamber and accessories shall be constructed and arranged as follows:

a. There shall be no direct impinging of the spray, or dripping of the condensate on the specimens.

b. The spray shall circulate freely about all specimens to the same degree.

c. No liquid, which has entered the exposure chamber and has come into contact with the test specimens, shall return to the salt-solution reservoir.

d. The chamber shall be properly vented.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:37Z
Check the source to verify that this is the current version before use.

4.1.2 Atomizers. The atomizer(s) used shall be of such design and construction as to produce a finely-divided, wet, dense fog.

4.1.3 Air supply. The compressed air entering the atomizer(s) shall be essentially free of all impurities, such as oil and dirt. Means shall be provided to humidify and warm the compressed air as required to meet the operating conditions. The air pressure shall be suitable to produce a finely-divided dense fog with the atomizer(s) used. To avoid any substantial difference between the salt concentration in the fog particles and that of the salt solution before atomization, the atomizing air must have a relative humidity between 95 and 98 percent. Thus, it is necessary to saturate the air at a higher temperature and cool it to 95°F (35°C) in the chamber. A satisfactory method is to pass the air in very fine bubbles through a tower containing water maintained at the proper temperature. Temperatures usually found to be satisfactory for various values of air pressure are given in table I.

TABLE I. Air pressure and temperature range.

Air pressure (lb/in^2)	Temperature	
	°F	°C
12	114	46
14	117	47
16	119	48
18	121	49

4.1.4 Salt solution. The salt used shall be substantially free of nickel and copper and containing on the dry basis not more than 0.01 percent sodium iodide and not more than 0.3 percent of total impurities. A 5-percent solution shall be prepared by dissolving 5 ±1 parts, by weight, of salt in 95 parts, by weight, of distilled water, or deionized water containing not more than 200 parts soluble solids per million. The solution shall be kept free from solids by filtration or decantation. The solution shall be adjusted to and maintained at a specific gravity of from 1.0268 to 1.0413 when measured at a temperature of 95°F (35°C).

4.1.4.1 The pH of the salt solution shall be so maintained that solution atomized at 95°F (35°C) will be in the pH range of 6.5 to 7.2. The pH measurement shall be made immediately after the collected sample is removed from the exposure chamber. The pH shall be measured electrometrically, using a glass electrode with a saturated potassium chloride bridge; or colorimetrically, using bromthymol blue as indicator. Only diluted analytical reagent grade hydrochloric acid or cp sodium hydroxide shall be used to adjust the pH.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:37Z
Check the source to verify that this is the current version before use.

4.1.4.2 When the pH of a salt solution is adjusted at room temperature and atomized at 95°F (35°C), the pH of the collected solution will be higher than that of the original solution due to the loss of carbon dioxide at the higher temperature. The following three methods have been found satisfactory for adjusting the pH of a sample collected after atomization at 95°F (35°C) within the pH range of 6.5 to 7.2.

a. Prepare the salt solution at room temperature and adjust the pH so that a 50-milliliter sample will have a pH between 6.5 and 7.2 after being boiled gently for 30 seconds and cooled to room temperature. The pH of the sample should be measured immediately after it is cooled. It is necessary for the pH of the original solution to be below 6.5 if this method is followed.

b. Heat the salt solution to boiling and cool to 95°F (35°C) or maintain the solution at 95°F (35°C) for approximately 48 hours before adjusting the pH to the required range of 6.5 to 7.2.

c. Heat the water from which the salt solution is to be prepared to 95°F (35°C) to expel the carbon dioxide before preparing the solution and maintain this temperature while adjusting the pH to within the range of 6.5 to 7.2.

4.2 Test specimens.

4.2.1 General. Specimens shall be given a minimum of handling, particularly on the significant surfaces, and shall be prepared for test immediately before exposure. Specimens should be handled with clean lintless gloves.

4.2.1.1 Unless otherwise specified, uncoated metallic specimens and metallic-coated specimens shall be thoroughly free of oil, dirt, and grease. Specimens shall be cleaned with detergent and rinsed with deionized water to obtain a water-break-free surface. The specimen shall then be rinsed with isopropyl alcohol and air or blow dried. No halogenated solvents or abrasives of any type are permitted. Cleaning must be done within one hour of test, and shall be processed with a minimum of handling. The specimen shall be handled only with clean, lintless cotton gloves after the cleaning operation.

4.2.2 The specimens will be racked or supported in such a manner that the spray will not condense and gather in recesses or other features of the fastener and will circulate freely about all specimens to the same degree.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:37Z
Check the source to verify that this is the current version before use.

5. DETAIL REQUIREMENTS

5.1 Test procedures.

5.1.1 Location of specimens. Whenever practicable, the specimen shall be supported from the bottom or the side. When specimens are suspended from the top, suspension shall be by means of nylon cord, glass hooks, or waxed string. The use of metal hooks is not permitted. Specimens shall be so positioned that:

a. They do not contact each other or any metallic material or any material capable of acting as a wick.

b. They are not shielded from the freely settling fog.

c. Corrosion products and condensate from one specimen does not fall upon another specimen or return to the solution reservoir.

5.2 Operating conditions.

5.2.1 Temperature. The test shall be conducted with a temperature in the exposure zone maintained at 95°F, +2°F, -3°F (35°C, +1°C, -1.7°C). Satisfactory methods for controlling the temperature accurately are by housing the apparatus in a properly controlled constant-temperature room, by thoroughly insulating the apparatus and preheating the air to the proper temperature prior to atomization, or by jacketing the apparatus and controlling the temperature of the water or of the air used in the jacket. No surface in the chamber shall become overheated enough to cause evaporation of the fog followed by condensation of that moisture on the other fog droplets, thereby diluting them. The use of immersion heaters within either (a) the chamber, for the purpose of maintaining the temperature within the exposure zone, or (b) in the salt-solution reservoir, is prohibited, except that suitable heating units may be used in larger exposure chambers, provided they do not cause the temperature within the exposure zone to vary outside the permissible range of operating temperature.

5.2.2 Atomization.

5.2.2.1 The conditions maintained in all parts of the exposure zone shall be such that a suitable receptacle placed at any point in the exposure zone will collect from 0.75 to 2.0 milliliter (ml) of solution per hour for each 80 square centimeters (sq cm) of horizontal collecting area (10 centimeter diameters (cm dia)) based on an average run of at least 16 hours. The solution thus collected shall have a sodium-chloride content from 4 to 6%, specific gravity (sp gr) of from 1.0268 to 1.0413 when measured at a temperature between 93°F and 97°F (34°C and 36°C) and a pH from 6.5 to 7.2. At least two clean fog-collecting

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:37Z
Check the source to verify that this is the current version before use.

receptacles shall be used; one placed as near to any nozzle and one as far from all nozzles as any specimen. Receptacles shall be so placed that they are not shielded from the freely settling fog and so that no drops of solution from specimens or from other sources will be collected.

5.2.2.2 Nozzles shall be made of material nonreactive to the salt solution. Glass or methyl-methacrylate (Lucite) nozzles with nylon filters are suggested. Suitable atomization has been obtained in boxes having a volume of less than 12 cubic feet with the following conditions:

a. Nozzle pressure of 12 to 16 pounds per square inch.

b. Orifices of from 0.02- to 0.03-inch in diameter.

c. Atomization of approximately 3 quarts of the salt solution per 10 cubic feet of box volume per 24 hours.

5.2.2.3 When using boxes having a volume in excess of 12 cubic feet, the above conditions may have to be modified in order to meet the requirements for operating conditions.

5.3 Length of test. The length of the salt spray test shall be 96 hours. Unless otherwise specified, the test shall be run continuously for the time indicated or until definite indication of failure is observed, with interruption only for adjustment of the apparatus and inspection of the specimen. Operations shall be so scheduled that interruptions are held to a minimum. When a retest is authorized for a sample(s) that failed to pass a specified test, the retest shall be run for the 96 hours with a minimum of interruptions or opening of the cabinet.

5.4 Examination of specimens. At the end of the test, the specimens shall be examined immediately to determine compliance with the standard of acceptability specified in the detail specification or drawing. If necessary to aid in examination, a gentle wash or dip in clean running water not warmer than 100°F (38°C) and blowing with clean dry air may be used. Examination at stated intervals during the test may be required.

6. NOTES

6.1 Test report. Unless otherwise prescribed in the detail specification or drawing for the material or product being tested, the test report shall include the following data:

a. Solution concentration.

b. Type of salt and water used in preparing the salt solution.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:37Z
Check the source to verify that this is the current version before use.

c. All readings of temperature within the exposure zone of the chamber.

d. Record of daily test made on each fog-collecting device, including the following:

1. Volume of salt solution collected in milliliters per hour per 80 sq cm of horizontal collecting area.

2. Sodium chloride content or sp gr (including temperature at which sp gr measurement was made) of the collected solution.

3. pH of collected solution.

e. Type of specimen and its dimensions, or number or description of product or part tested.

f. Method of supporting specimen in the salt spray chamber.

g. Description of protection used on cut edges and other surfaces not required to be tested.

h. Length of exposure period.

i. Cause and duration of any interruptions during test.

j. Results of all inspections.

k. Method of specimen surface preparation.

NOTE: For quality assurance testing, the report shall only specify:

a. That the test was conducted in accordance with MIL-STD-1312-1;

b. The exposure time; and

c. The results of final inspection reported.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:37Z
Check the source to verify that this is the current version before use.

Custodians:
Army - AV
Navy - AS
Air Force - 11

Review activities:
Army - AV, AR
Navy - AS, SH
Air Force - 11
DLA - IS

Preparing activity:
Navy - AS
(Project No. 53GP-0143-01)

☆ U.S. GOVERNMENT PRINTING OFFICE: 1984-505-038/A5112

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:37Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY
Commanding Officer
Naval Air Engineering Center
Engineering Specifications and Standards Department
(ESSD), Code 93
Lakehurst, NJ 08733

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12503 WASHINGTON D C

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commanding Officer
Naval Air Engineering Center
Engineering Specifications and Standards Dept.
(ESSD), Code 93
Lakehurst, NJ 08733

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:37Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-STD-1312-1	Fastener Test Methods - Method 1, Salt Spray

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T23:37Z
Check the source to verify that this is the current version before use.